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A systematic review of green supply chain network design literature focusing on carbon policy

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ABSTRACT

Developed countries are responsible for a great share of the historical carbon dioxide (CO_2) emissions in the atmosphere. The need to reduce CO_2 emissions is now widely accepted, but how this can be accomplished is still controversial and debated. Considering the growing size of the carbon trading market and the enlargement of carbon business mechanisms, CO_2 should be considered when designing supply chain networks (SCNs). Many researchers are developing green and sustainable supply chain networks to diminish greenhouse gas release. This paper reviews the literature on green supply chain network design (GSCND) from 2010 to 2023 and analyzes the effect of related carbon regulations on GSCND. We focus on methodologies and models that expressly account for carbon releases. We show three policies that have influenced the design of SCNs on carbon emissions: carbon cap, cap-and-trade, and carbon tax.

1. Introduction

Carbon reduction has become one of the most important objectives in designing supply chains (SCs) due to environmental conservation, increasing customer awareness, and stringent carbon policies. Our goal is to answer the challenges ahead by policymakers and SC decision-makers by reviewing recent literature on green supply chain network design (GSCND) and analyzing how emissions, carbon policies, and their ramifications on the SC are measured and modeled. Most policymakers aim to design effective carbon policies, diminish greenhouse gas (GHG), and establish efficient procedures for auditing and certifying these facilities and equipment. Multiple industries are interested in the financial impact of policies and ways of responding to them. To reduce carbon emissions, companies must restructure their designing supply chain networks (SCNs), participate in a low-carbon system, and practice energy efficiency [1].

The global carbon trading market has grown explosively since the Kyoto Protocol came into force in 2005. Globally, many countries have introduced policies aimed at reducing GHG emissions. Globally, 1/4 of GHG emissions come from 25 subnational and 42 national governments, according to the World Bank [2].

Due to the relatively late start of China's carbon trading market, which follows the EU's quota trading mechanism, the carbon policy has many deficiencies and should be improved. Therefore, reducing carbon emissions is critical for supply chain network design (SCND) [3]. To achieve economic efficiency and environmental benefit, GSCND is an emerging approach. About one-third of SC emissions are caused by transportation. Followed by raw material procurement, facility construction, and distribution center operation [1]. Under cap-and-trade regulations, SC coordination with green technology is essential [4].

A carbon policy plays a crucial role in reducing carbon emissions in green supply chains (GSCs). A carbon tax, carbon caps, and cap-and-trade, also carbon subsidies, are generally considered by the government after reading these studies. A carbon tax, carbon cap, and carbon trading are the three dimensions mainly studied in GSCND [5].

The remaining four sections are presented in this paper. In Section 2, a detailed description of the carbon policy is provided. The finding and filling of the research gap are discussed in Section 3. The Managerial implications and practical insights are shown in Section 4. The conclusion of this paper is presented in Section 5, also suggestions for future research.

2. Carbon policy

In designing SCNs, carbon policies have complex and significant implications. We reported SC networks considering carbon policies. The carbon policy of SCND researchers between 2010 and 2023 is summarized in this report. In addition to carbon taxes, cap-and-trade policies are also in use. Our research focuses on supply chain design (SCD) work that accounts for carbon regulations. We are particularly interested in a paper that presents models, conducts numerical analyses

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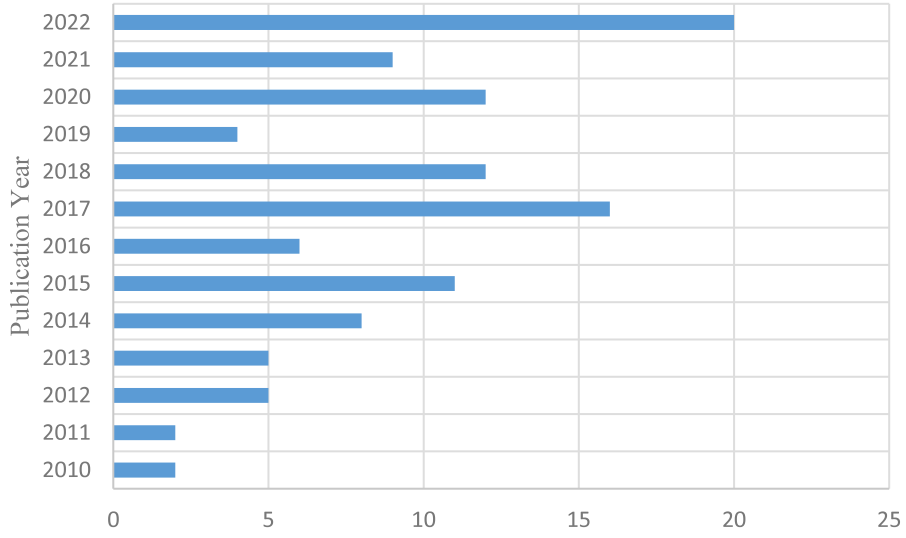

Fig. 3. Publication year of papers.

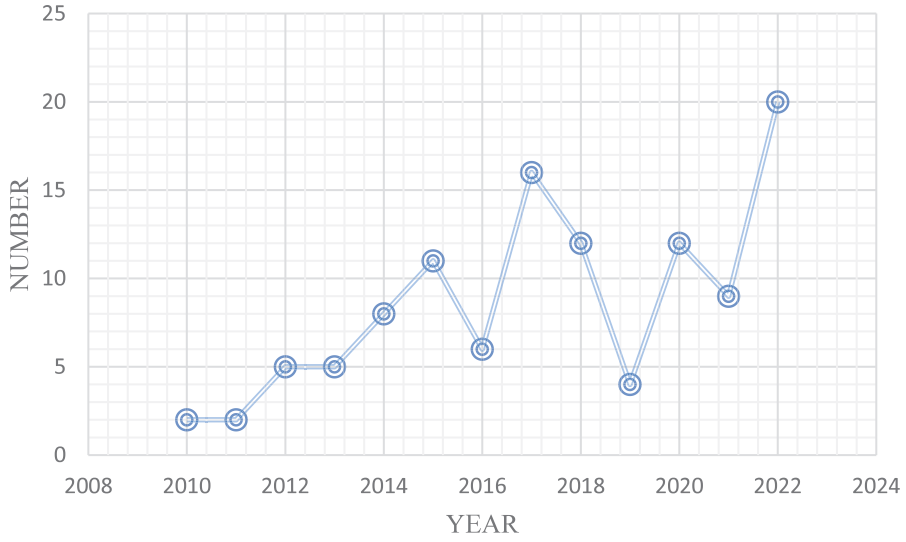

Fig. 4. Distribution of papers over time.

cap subject is shown according to the network analysis. The network analysis in Fig. 13 is based on the cap-and-trade subject from 2010 to 2023. The artificial intelligence analysis shows that carbon dioxide emissions and related policies are among the most important issues for the green chain. The importance of writing a good review paper is significant at this stage.

The research conducted in the field of carbon between 2010 and 2023 is reviewed in this section. In Figs. 1, 2 and 3 research results about carbon policy, carbon tax, and carbon cap are presented in a grid, with circles representing the areas where most research has been conducted over the past few years. Their sizes correspond to how much research has been conducted in these areas.

There is a direct connection between the two fields. It depends on the weight of an item and how big the label and circle will be. A larger label and circle are associated with a heavier item. A larger circle indicates more research has been conducted in that area. Differentiated clusters with different colors indicate close relationships between them. These clusters are connected by lines. Using these lines, we can see how many research fields are connected to each on the evaluation results, clusters in different classifications are linked in a variety of ways.

In contrast to the carbon tax, carbon cap, and carbon policy, carbon policy has 5, 39, and 9 clusters, respectively. In this section, the

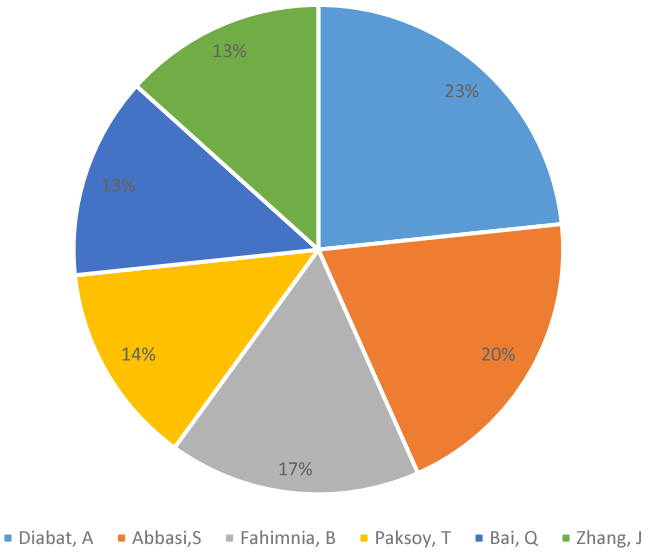

Fig. 5. Objective description of leading authors.

Table 1
An overview of carbon policy for GSCNC researchers between 2010 and 2023.

Author(s) and references	Year	Focused carbon tax	Focused carbon cap	Focused Cap-and-Trade
Abdallah et al. [85]	(2010)			*
Paksoy [93]	(2010)			*
Chaabane et al. [57]	(2011)			*
Paksoy et al. [78]	(2011)	*		
Akgul et al. [94]	(2012)	*		*
Abdallah et al. [95]	(2012)			*
Chaabane et al. [86]	(2012)			*
Giarola et al. [81]	(2012)			*
Kannan et al. [58]	(2012)			*
Du et al. [64]	(2013)			*
Abdallah et al. [61]	(2013)			*
Diabat, A et al. [59]	(2013)			*
Benjaafar et al. [20]	(2013)		*	
Mirzapour Al-e-Hashem et al. [15]	(2013)		*	
Baud-Lavigne et al. [54]	(2014)		*	
He et al. [96]	(2014)	*	*	
Jin et al. [97]	(2014)	*	*	*
Paksoy & Özceylan [98]	(2014)	*		
Marufuzzaman et al. [11]	(2014)	*	*	*
Zeballos et al. [99]	(2014)	*		
Oh and Jeong [51]	(2014)		*	
Zhang et al. [21]	(2014)	*		
Choudhary et al. [16]	(2015)	*	*	*
Fahimnia et al. [9]	(2015, a)	*		
Du et al. [67]	(2015)			*
Fareeduddin et al. [17]	(2015)	*	*	*
Fahimnia et al. [100]	(2015, b)			
Liotta et al. [101]	(2015)	*		
Martí et al. [50]	(2015)	*	*	
Niakan et al. [102]	(2015)	*		
Rezaee et al. [48]	(2015)		*	*
Zakeri et al. [10]	(2015)	*		*
Hammami et al. [22]	(2015)	*	*	
Tao et al. [23]	(2015)		*	
Alhaj et al. [103]	(2016)	*		
Xu et al. [68]	(2016)			*
Liotta et al. [104]	(2016)	*	*	
Yang & Yu [105]	(2016)	*		
Peng et al. [12]	(2016)	*	*	
Ding et al. [25]	(2016)		*	
Golpîra et al. [77]	(2017)			*
Xu et al. [24]	(2017)	*		*
Xu et al. [47]	(2017)			*
Arampantzi & Minis i et al. [80]	(2017)			*
Entezaminia et al. [62]	(2017)	*	*	
Mohammed et al. [18]	(2017)	*	*	*
Hariga et al. [106]	(2017)	*		
Soleimani et al. [19]	(2017)		*	
Zhou et al. [14]	(2017)	*		
Liu et al. [107]	(2017)			*
Qi et al. [27]	(2017)			*
Wang et al. [108]	(2017)	*		
Li et al. [60]	(2017)			*
Ji et al. [29]	(2017)			*
Yu & Han [109]	(2017)	*		
Li et al. [110]	(2017)	*		
Shu et al. [111]	(2018)	*		
Yi et al. [112]	(2018)	*		
Ma et al. [113]	(2018)	*		
Haddadsisakht& Ryan [114]	(2018)	*		
Olsen et al. [115]	(2018)	*		
Guo et al. [116]	(2018)	*		
Zhou et al. [117]	(2018)	*		
Xu et al. [63]	(2018)			*
Bai et al. [71]	(2018)			*
Xia et al. [34]	(2018)			*
Wang et al. [38]	(2018)			*
Wang et al. [31]	(2018)			*
Tong et al. [73]	(2019)			*
Wong et al. [118]	(2019)	*		
Waltho et al. [1]	(2019)	*	*	*
Zhang et al. [28]	(2019)		*	

(*continued on next page*)

Table 1 (*continued*).

Author(s) and references	Year	Focused carbon tax	Focused carbon cap	Focused Cap-and-Trade
Cadavid-Giraldo et al. [30]	(2020)	*	*	
Kuiti et al. [72]	(2020)	*		
Cao et al. [119]	(2020)	*		
Babagolzadeh et al. [120]	(2020)	*		
Bai et al. [121]	(2020)	*		*
Zou et al. [122]	(2020)	*		
Kushwaha et al. [32]	(2020, a)			*
Ghosh et al. [36]	(2020, b)			*
Ghosh et al. [37]	(2020)			*
Mishra et al. [40]	(2020)			*
Tang & Yang [41]	(2020)			*
Yang et al. [56]	(2020)			*
Tong et al. [74]	(2019)			*
Taleizadeh et al. [70]	(2021)			*
Xu, & Choi [65]	(2021)			*
Li et al. [66]	(2021)			*
Yang et al. [39]	(2021)			*
Wang&Wu [35]	(2021)			*
Lang et al. [123]	(2021)	*		
Liu et al. [124]	(2021)	*		
Zhang et al. [125]	(2021)	*		
Guo & Xi [126]	(2022)	*		
Meng et al. [127]	(2022)	*		*
Zhu et al. [128]	(2022)	*		
Hu, & Wang [129]	(2022)	*		
Zhang & Zhang [130]	(2022)	*		
Paul et al. [131]	(2022)	*		
Shi& Liu [132]	(2022)	*		
Shen et al. [133]	(2022)	*		*
Wu et al. [134]	(2022)	*		
Yu et al. [135]	(2022)	*		
Lyu et al. [136]	(2022)	*		
Cheng et al. [137]	(2022)	*		*
Luo et al. [138]	(2022)	*		
Xu et al. [139]	(2022)	*		
Astanti et al. [33]	(2022)			*
Wang et al. [69]	(2022)			*
Chen et al. [140]	(2022)	*		
Wei & Huang [141]	(2022)	*		
Luo et al. [142]	(2022)	*		
Yi et al. [143]	(2022)	*		
Abbasi et al. [144]	(2023)	*	*	*

research conducted from 2010 to 2023 in the field of carbon has been reviewed. Figs. 2, 1, 3, and 4 respectively show the results of the research conducted in the field of Carbon Policy, Carbon Tax, and Carbon Cap in the form of a grid, where each circle represents the most frequent areas in recent years, and the size of these circles corresponds to the amount of research conducted in them. The field is directly related. The size of the label and circle of an item is determined by the weight of that item.

The bigger the weight of an item, the bigger the label and circle. Circles with larger sizes indicate a high number of research conducted in that research field. Differentiated clusters with different colors indicate their close relationship to each other, which are connected through lines. These lines show the number of connections of each research field with other fields. The evaluation results indicate strong and diverse links between clusters in different classifications. A carbon tax with 70 items and a carbon cap with 62 items have the most and least communication items.

Figs. 5, 6, 7 and 8 show where research has been pushed in the areas of carbon policy, carbon tax, carbon cap, and cap-and-trade during the period 2010–2023 and in what areas there has been more activity. Social welfare, carbon tax mechanisms, and supply chain have been gaining more attention in recent years in the area of carbon tax. As well, recent years have seen more attention paid to policy recommendations and trade regulation in the fields of carbon Polices and carbon cap, and more research has been conducted in these fields, as well as the cap-and-trade issue claimed.

3. Finding and filling the research gap

The general finding and research innovations include the following:

- An examination of green supply chain network design based on carbon policy is presented in this study.
- Our study analyzes the impact of carbon regulations over a decade on green supply chain network design.
- A consumer's attitude toward CO_2 emissions is taken into account.
- Based on the keywords carbon policy, carbon tax, and carbon cap, we present several network design diagrams.
- Policies such as carbon caps, cap-and-trade, and carbon taxes influence the design of supply chain networks.

4. Managerial implications and practical insights

This is a literature review and comprehensive paper about GSCND focusing on the carbon rule that can solve all the information needs of managers in this field. Most of the essential articles from 2010 to 2023 have been included in this manuscript. As a result of our study, we believe the following managerial implications can be drawn:

1. For carbon policies to be considered in SC design, all parties involved in the finance section must understand the critical factors that influence this design. As a result of this study, they can focus on improving the adoption and effectiveness of SC by concentrating on these factors.

Table 2
The distribution of articles based on the source of publication.

Name of the journal	References	No. of articles
Journal of cleaner production	[24], [80], [27], [112], [106], [97], [125], [124], [117], [32] [30], [108] , [63], [68], [63], [68]	17
Sustainability	[111] , [112], [134], [109], [122], [121], [38], [70], [83], [70], [82], [51]	13
International Journal of Production Economics	[1], [86], [61], [16], [4], [138], [100], [114], [25], [72], [29], [101], [50], [10], [22]	15
Transportation Research Part E: Logistics and Transportation Review	[39], [41], [69], [65], [8]	5
Transportation Research Part D: Transport and Environment	[120], [145], [18]	3
Computers & Chemical Engineering	[94], [99]	2
Flexible Services and Manufacturing Journal	[84], [52]	2
Energy Economics	[66], [146]	2
Discrete Dynamics in Nature and Society	[96], [21]	2
International Journal of Production Research	[31], [35]	2
Resources, conservation, and recycling	[58], [23], [103], [73], [74]	4
Journal of the Operational Research Society	[98], [119]	2
Transportation Research Part B: Methodological	[147]	1
Production Planning & Control	[57]	1
Scientia Iranica	[77]	1
Transportation Science	[11]	1
Climate Policy	[148]	1
Economic and social review	[149]	1
Energy Policy	[64]	1
International Journal of Internet Manufacturing and Services	[116]	1
International Journal of Economics, Finance and Management Sciences	[142]	1
Benchmarking: An International Journal	[36]	1
Technology, Market, and Complexity	[33]	1
Mathematics	[40]	1
Applied Mathematical Modeling	[95]	1
IEEE Transactions on engineering Management	[59]	1
IEEE Transactions on automation science and engineering	[20], [14]	2
European Journal of Operational Research	[15], [56]	2
Scientific Research and Essays	[93]	1
Environmental Science and Pollution Research	[130]	1
Dynamic Games and Applications	[129]	1
International Transactions on Electrical Energy Systems	[127]	1
Mobile Information Systems	[140]	1
Bioresource technology	[81]	1
Journal of Chinese Economic and Foreign Trade Studies	[105]	1
Decision Sciences	[113]	1
Folia Oeconomica Stetinensia	[150]	1
Procedia	[17]	1
Engineering Optimization	[102]	1
Annals of operations research	[48], [131], [34], [67]	4
IEEE Transactions on Industrial Informatics	[104]	1
Mathematical Problems in Engineering	[132], [128], [12]	3
The International Journal of Advanced Manufacturing Technology	[62]	1
Computers & industrial engineering	[135], [121] [143] [28] [71] [74], [151] [79], [19]	9
Open Physics	[60], [110]	2
IEEE Transactions on Power Systems	[115]	1
Energy & Environment	[118]	1
Omega	[139], [47]	2
Sustainable Production and Consumption	[137]	1
International Journal of Environmental Research and Public Health	[136], [133], [107], [141]	4
IOP Publishing	[123]	1

2. We have identified and covered different policies and ways to solve the prices of CO_2 emissions, which can help create awareness for the manager's supply chain.

3. Highlights useful information about this paper and the paper we discuss providers may also benefit from managers and any who involve in the SC process.

5. Conclusion, limitations, and future works

5.1. Conclusion

In comparison to carbon taxes, carbon caps, and cap-and-trade, carbon taxes exert pressure from economics on enterprises, resulting in a more effective policy. These studies come to numerous common conclusions. The costs cannot rise sharply by imposing a carbon tax on the SC. Furthermore, to decrease CO_2 and waste resources, Wong et al. assert that a carbon tax can promote a more comprehensive carbon policy reform. As the range of a carbon tax is small, setting a standard is prudent. As opposed to carbon caps with carbon taxes, carbon tax policies are usually inflexible, but they can contribute to lower overall costs. SC impacts are evident when carbon caps are raised. As a consequence of the carbon cap, emissions sharply reduce, while costs only increase slightly. The product is recycled and reused in the forward, reverse, and CLSCs due to the application of optimal production technology. Similarly, to carbon caps, the higher the cap in cap-and-trade, the lower the cost in the SC. With a cap-and-trade policy,

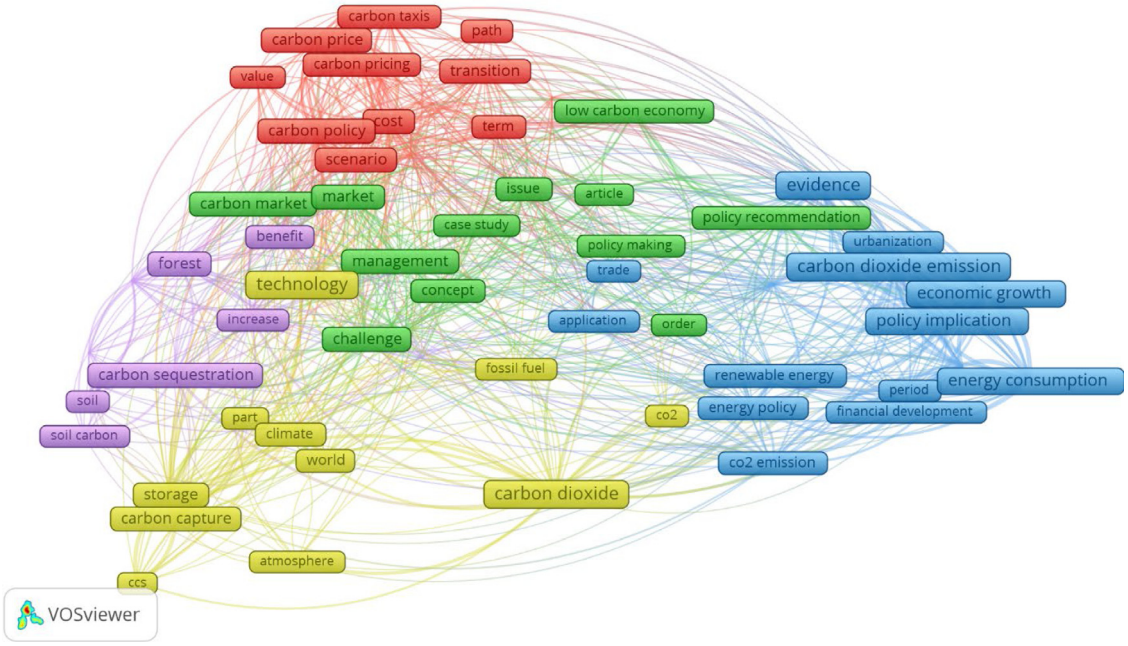

Fig. 6. Network analysis based on a carbon policy keyword diagram.

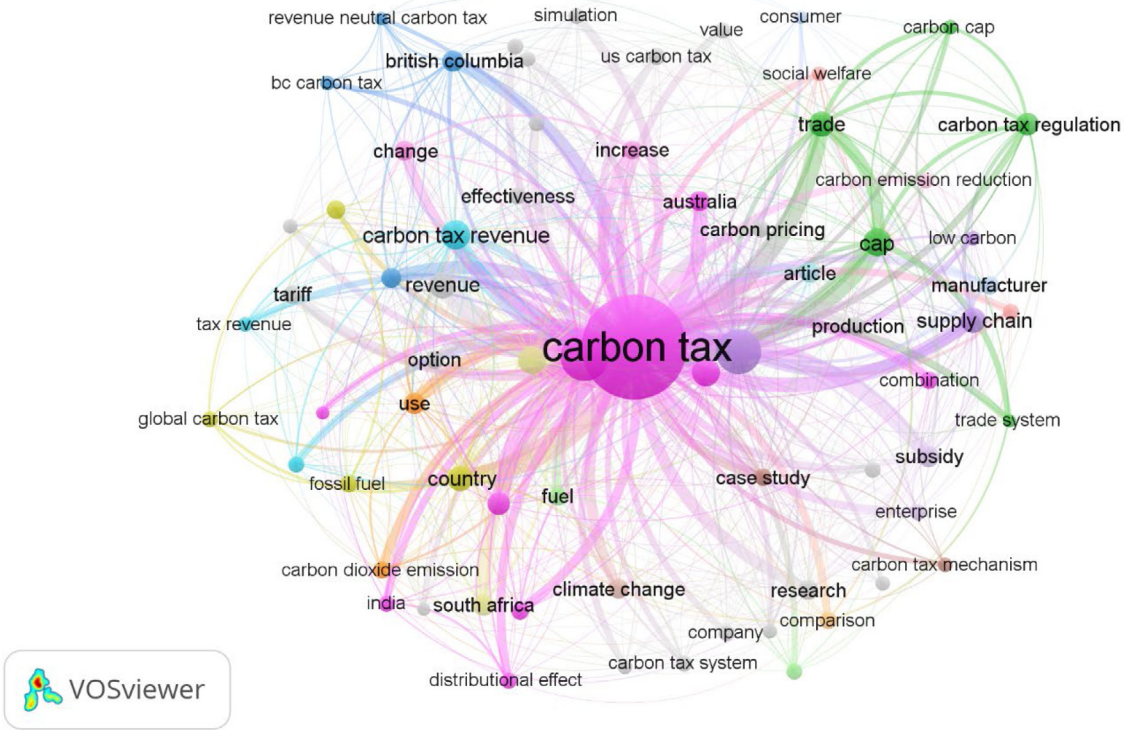

Fig. 7. Network analysis based on a carbon tax keyword diagram.

carbon emissions and total costs could be reduced simultaneously. To encourage the industry to pursue GSCND, cap-and-trade policies are beneficial. Furthermore, GSCND is dependent upon a carbon credit. This article aims to present a review of the GSCND research. Many articles analyze carbon policies, and carbon taxes, caps, and cap-and-trade are commonly utilized. Since 2010, there has been an increase in papers because environmental concerns are constantly arising. In addition to carbon tax policy, GSCD, and strategy formulation also play a significant role. Carbon emissions cost different policies can minimize rising. To reduce emissions in a CLSC, most research focuses on transportation modes, facility locations, manufacturing, electricity, and suppliers. A review of articles published between 2010 and 2022 was conducted to examine themes that inexplicably studied green, friendly to the environment, and sustainable supply chain network design (SSCND) and modeled the trade-offs between cost and environmental effects. SCND is analyzed in terms of carbon rule and quantifiable resale

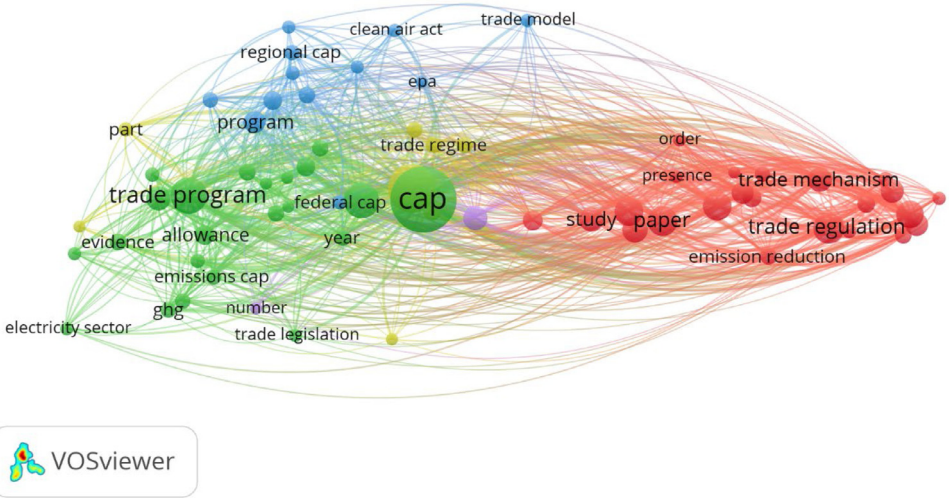

Fig. 8. Network analysis based on a carbon cap keyword diagram.

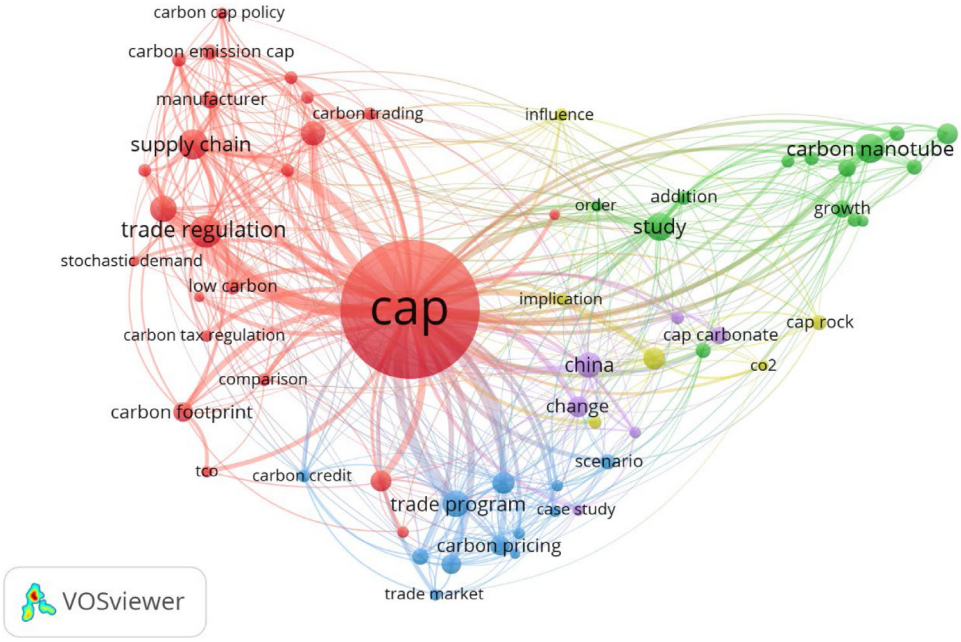

Fig. 9. Network analysis based on a cap-and-trade keyword diagram.

Table 3
An analysis of the distribution of papers by time.

Year	Total number
2010	2
2011	2
2012	5
2013	5
2014	8
2015	11
2016	6
2017	16
2018	12
2019	4
2020	12
2021	9
2022	20

Table 4
This field's leading authors.

Author	Number of contributions	Timeline
Diabat, A	7	2009–2012
Abbasi, S	6	2021–2023
Fahimnia, B	5	2015–2021
Paksoy, T	4	2010–2014
Bai, Q	4	2016–2018
Zhang, J	4	2015–2019

sources. We examine the four standard carbon cap, cap-and-trade, and carbon tax practices in detail, concentrating on their usage and impact on the SC. In the right circumstances, carbon taxes can reduce emissions without increasing costs significantly. It is also possible to achieve desirable reductions through carbon caps and cap-and-trade rules with a little increase in price. Due to its flexibility, cap-and-trade is the most favored of the four policies. Investing in green technology and reducing emissions is further facilitated by the ability to sell off unused credits. We then analyze how SCND captures Sources of varying SC emissions.

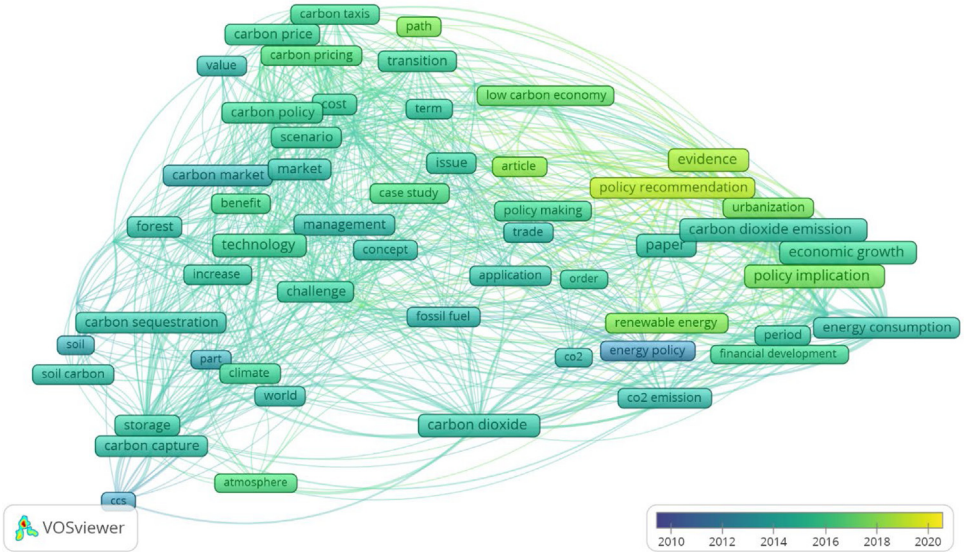

Fig. 10. Network analysis according to the carbon policy subject in 2010–2022.

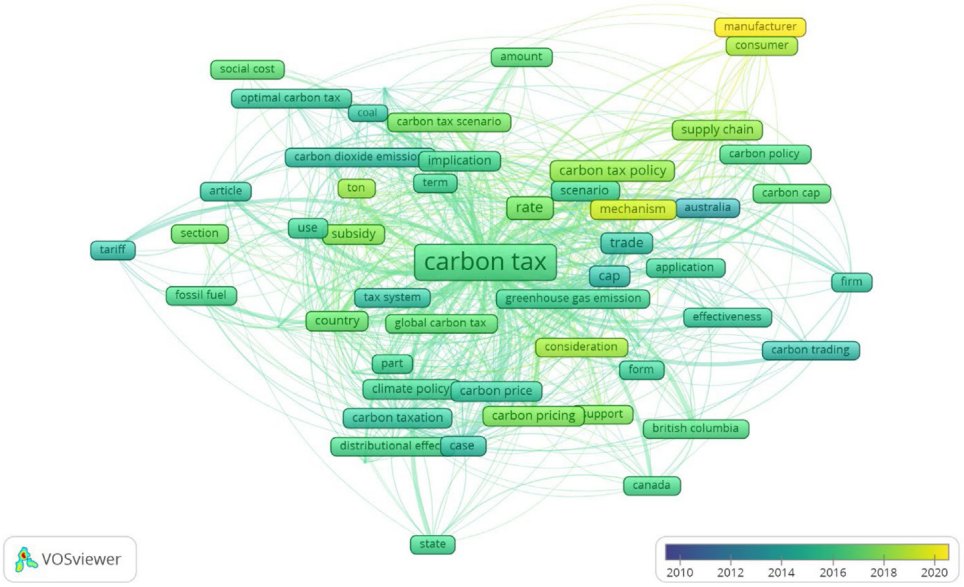

Fig. 11. Network analysis according to the carbon tax subject in 2010–2022.

Most emission models assume that emissions are linear functions of a single variable; in other words, it is a simplistic view in practice. It would be better if emissions were represented more accurately, perhaps in a nonlinear and multivariate way. In the same way, carbon policies have the same effect. It would be better not to apply a blanket cap, credit price, or tax rate to all emissions, but rather to them according to their source. Research suggests that small cost increases can result in substantial emissions reductions. To do this, resources must be optimally used, and materials must flow efficiently throughout the SC. According to the review, consumers and decision-makers of SC are becoming increasingly aware of the environment, resulting in a sample shift in SCN design. The carbon footprint will influence market share and SC configuration, accumulating as products move through the SC. Also, environmental rules are being implemented globally, which will affect the sourcing, producing, storing, and disposing of products. In response to consumers' increasing awareness of climate change and concern, emission-sensitive demand must be considered.

5.2. Limitations

The limitations of this research that can be addressed in future research include the following:

1. Data and information are lacking, resulting in problems that make it difficult for researchers to access books, journals, statistics, and databases.
2. Due to time and cost limitations, more cited articles can be added.
3. It can be the most productive and influential authors, institutions, and countries with the new countries' research.

5.3. Future works

Investing more time and effort into understanding how knowledge accumulates within the supply chain could be a worthwhile endeavor. Researchers could gain new insights into improving supply chain performance by conducting more research in this area. The fragmented

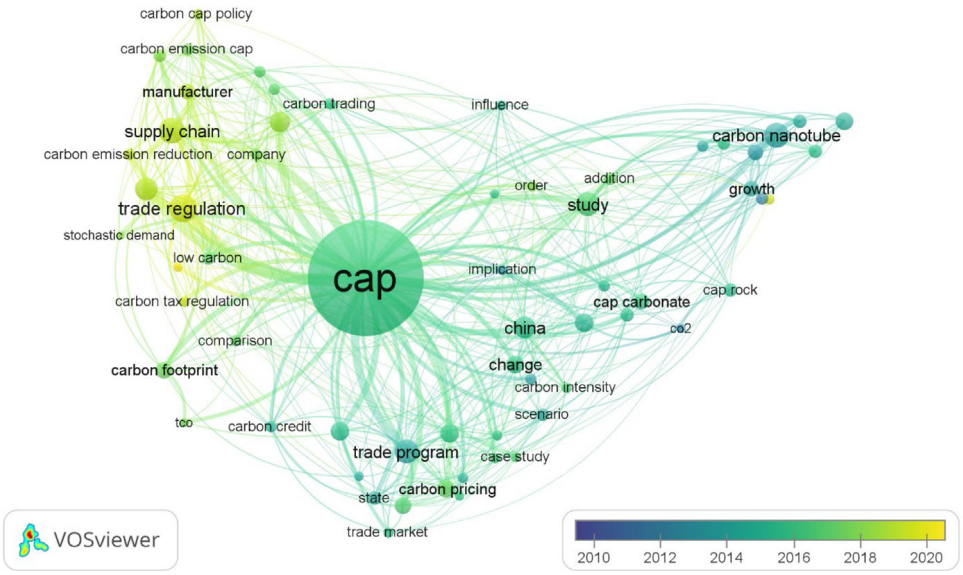

Fig. 12. Network analysis according to the carbon cap subject in 2010–2022.

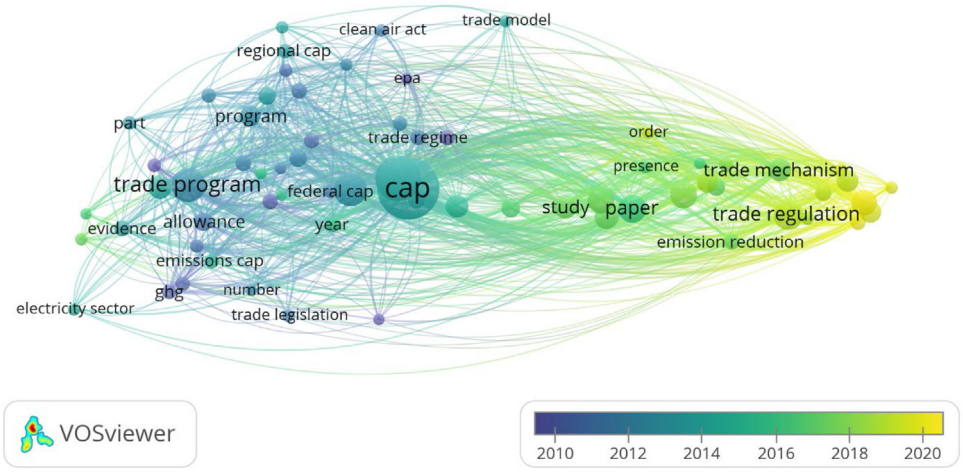

Fig. 13. Network analysis according to the cap-and-trade subject in 2010–2022.

nature of complex SCs is also associated with knowledge obsolescence, as discussed in many papers. The lack of understanding on this topic is another issue that needs to be addressed. Thus, to achieve superior performance at the firm level, it is necessary to understand the role of knowledge within the supply chain. Considering other policies such as carbon offset. A comprehensive review of international carbon policies on a GSCND is lacking. More papers before 2010 can also be reviewed. Research in this area investigation is warranted in the future. In addition, increasing the number of articles allows for a more comprehensive analysis of specific topics and a more accurate identification of trends. You might need to expand the search with new related keywords if the keywords used are not all-inclusive.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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